

ASSIGNMENT

TABLEAU BASICS

Q1. Explain what Tableau is and why it is preferred over traditional spreadsheet tools for data analysis. Also mention any two industries where Tableau is commonly used and explain one use case.

ANS- Tableau is a software tool used for **data visualization and analysis**.

- It helps turn raw data into interactive charts, graphs, and dashboards.
- Unlike spreadsheets (like Excel), Tableau is faster with large datasets, easier to use with drag-and-drop features, and gives dynamic visuals instead of static tables.

Why It's Better Than Spreadsheets

- **Handles Big Data:** Works smoothly with millions of rows.
- **Interactive Dashboards:** You can filter, drill down, and explore data visually.
- **Less Manual Work:** No need for complex formulas—just drag and drop.
- **Collaboration:** Dashboards can be shared online easily.

Industries Using Tableau

1. **Retail & E-commerce**
 - Use case: Analyzing sales trends by product category and region.
 - Example: A store can see which products sell best in which cities and adjust inventory.
2. **Healthcare**
 - Use case: Monitoring patient outcomes and hospital resources.
 - Example: Hospitals track bed occupancy and treatment success rates to plan resources better.

Q2. After connecting to the Sample – Superstore dataset: a) List the main types of fields available (examples: time-based, geographical, numerical). b) Identify any four Dimensions and any four Measures from the dataset.

ANS- Types of Fields in Tableau (Sample – Superstore)

When you connect to the dataset, Tableau organizes fields into two main categories:

- Time-based fields → e.g., *Order Date*, *Ship Date*
- Geographical fields → e.g., *Country*, *State*, *City*, *Postal Code*
- Numerical fields → e.g., *Sales*, *Profit*, *Quantity*, *Discount*

These appear in the Data Pane on the left side, with blue (Dimensions) and green (Measures) indicators.

Dimensions (Qualitative fields – Blue)

Examples from Sample Superstore:

1. Category
2. Sub-Category
3. Region
4. Customer Name

Dimensions are used to slice and group data (like categories, names, or locations).

Measures (Quantitative fields – Green)

Examples from Sample Superstore:

1. Sales
2. Profit
3. Quantity
4. Discount

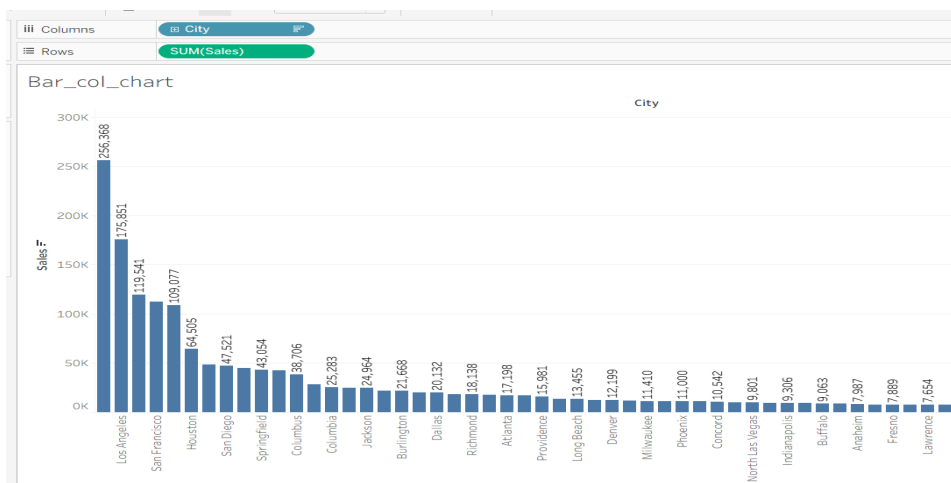
Measures are numeric values that can be aggregated (sum, average, min, max).

Q3. Create a Bar Chart showing Total Sales by Category. Explain: Which field goes to Rows Which field goes to Columns Which aggregation is applied to Sales

ANS- Rows → Sales (SUM)

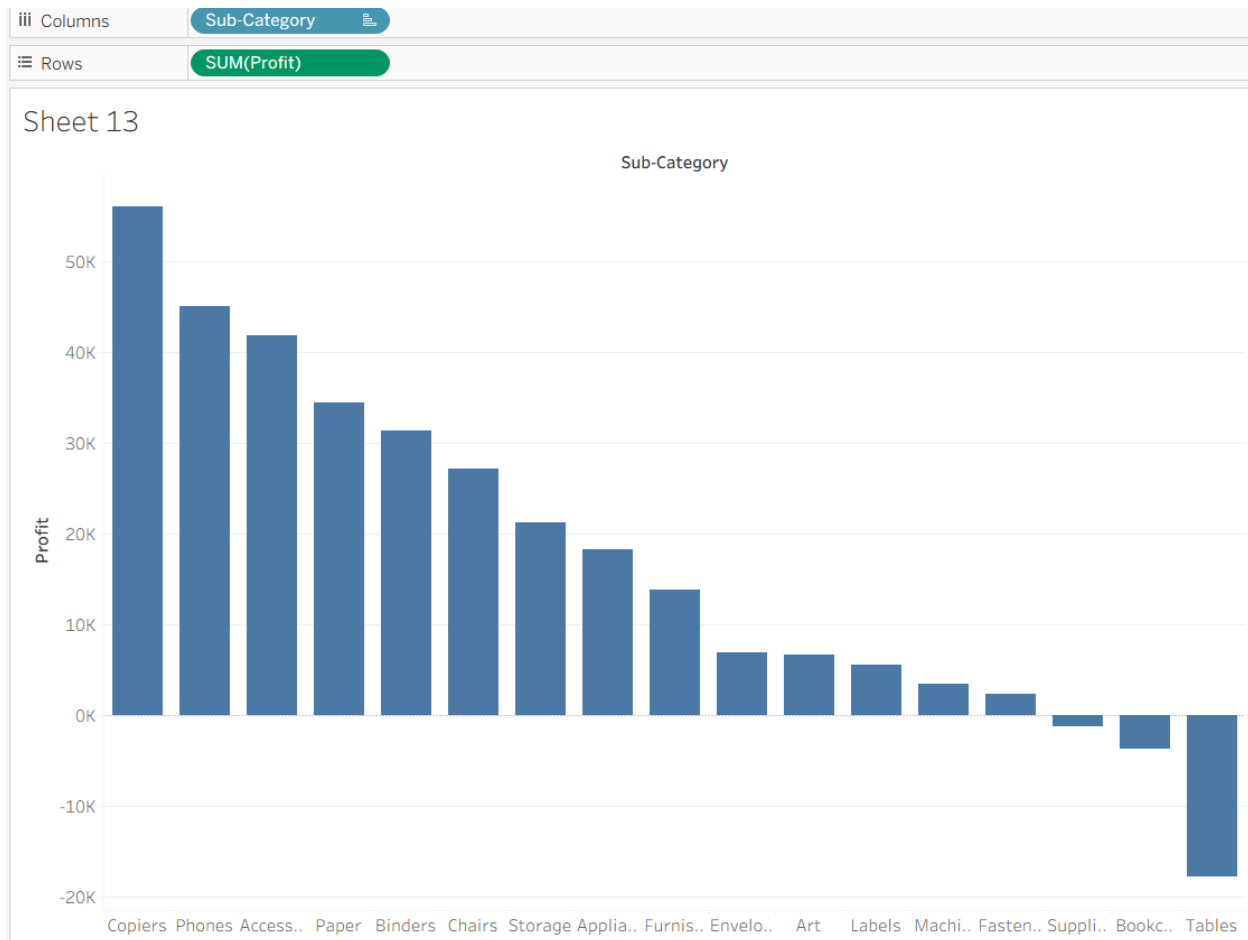
Columns → Category

Tableau automatically aggregates Sales using SUM, so the bar chart shows Total Sales by Category.



Q4. What insight can be gained by comparing Sales and Profit together? Then explain how you would create a chart to analyze Profit by Sub-Category. Hint: Use Sub-Category as a Dimension and Profit as a Measure

ANS-

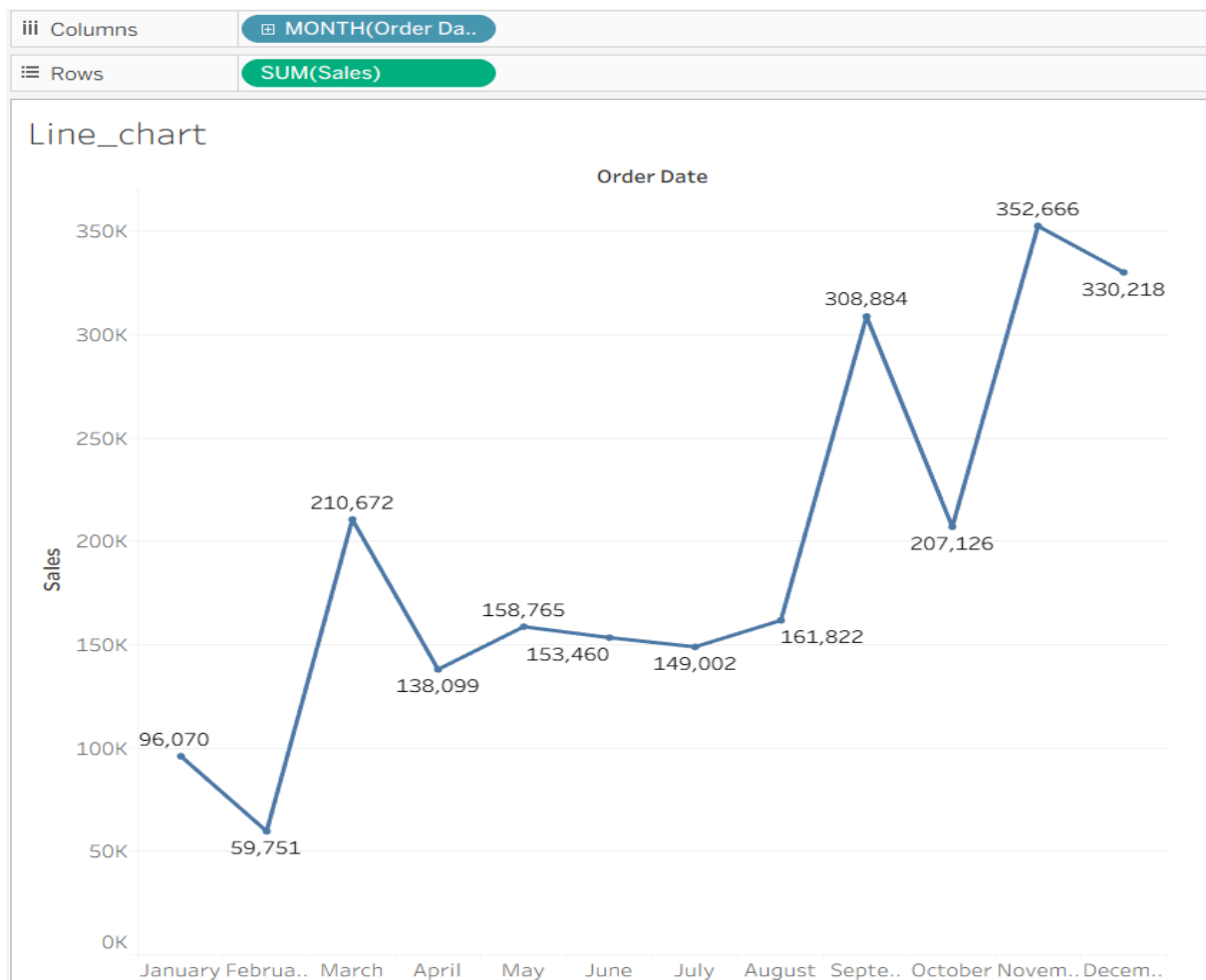


Insight from Comparing Sales and Profit:

- By analyzing **Sales and Profit together**, you can see which products or categories **generate high revenue but low profit**.
- This helps identify areas where **discounts or costs** may be reducing profitability.
- Example: A category might have high sales but low profit, showing it's popular but not very profitable.

Q5. Create a Line Chart to show Sales Trend over Order Date. Which date level would you choose (Year / Quarter / Month)? Why are line charts suitable for time-based analysis?

ANS-



Which Date Level to Choose?

- The most common choice is Month.
- Reason: Month-level detail shows clear seasonal or monthly sales patterns, while still being easy to interpret.

Why Line Charts for Time-Based Analysis?

- Line charts are ideal because they show continuous trends over time.
- The connected points make it easy to see growth, decline, and seasonal fluctuations.
- They highlight patterns such as peaks, dips, or long-term upward/downward trends.

Q6. Apply a Basic Filter to display data only for: Region = “West” Category = “Technology” Explain the steps and mention which shelf is used.

ANs- Steps in Tableau:

1. Connect to the Sample – Superstore dataset.

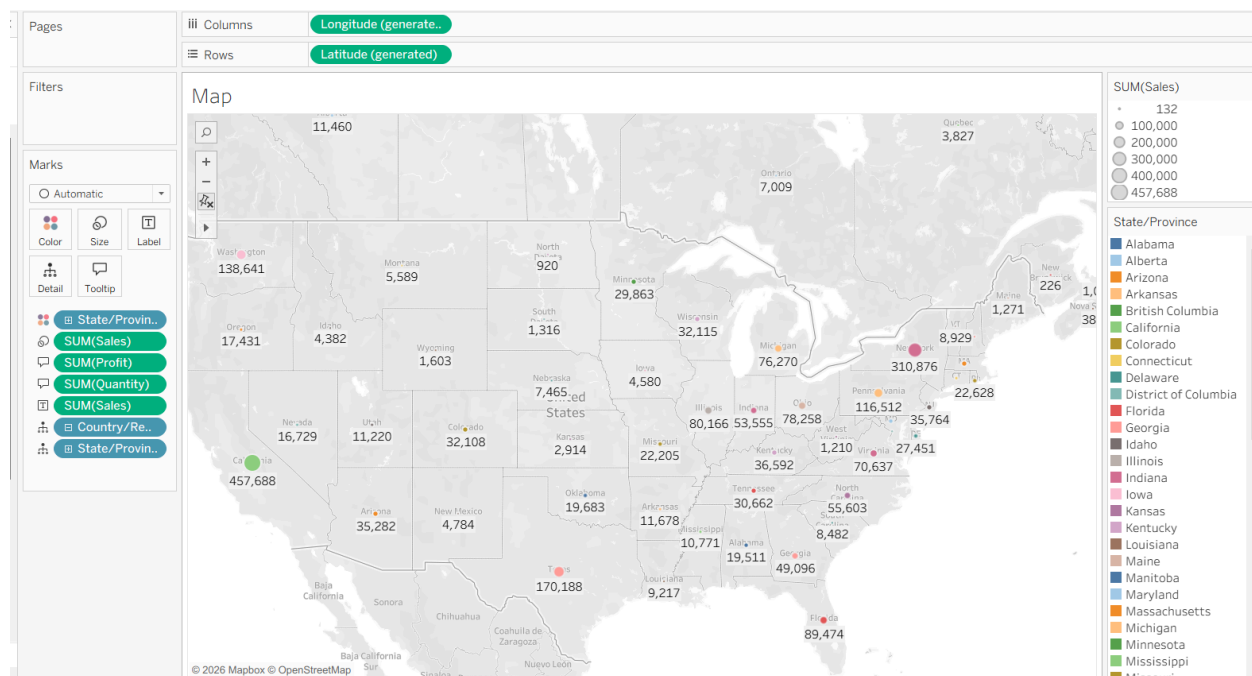
- In the Data Pane, drag the field Region to the Filters Shelf.
 - Select only “West” from the filter options.
- Similarly, drag the field Category to the Filters Shelf.
 - Select only “Technology”.
- Tableau will now display data restricted to West region and Technology category.

Which Shelf is Used?

- The Filters Shelf is used to apply these conditions.

Q7. Explain the business importance of geographical analysis. Then describe how you would create a Map showing Sales by State using Sample – Superstore.

ANS- Business Importance: Geographical analysis helps businesses see where their sales or profits come from. It identifies high-performing regions and areas needing improvement, supporting better marketing and supply decisions.



Q8. What is the difference between a Worksheet and a Dashboard in Tableau? Why is a dashboard more useful for decision-makers?

ANS- Worksheet:

- A single view in Tableau (like one chart or graph).
- Focuses on analyzing data from one perspective at a time.

Dashboard:

- A collection of multiple worksheets combined into one screen.
- Allows you to see different charts together, interact with filters, and compare insights side by side.

Why Dashboards are More Useful for Decision-Makers:

- Dashboards provide a holistic view of business performance.
- They combine multiple KPIs (sales, profit, trends, geography) in one place.
- Decision-makers can interact with filters and quickly spot patterns, making it easier to take action.

Q9. What is Tableau Prep, and why is it important in real-world data projects? Then explain how you would: Remove null values Remove duplicate records Change incorrect data types (Answer in steps, no tool access required).

ANS - Tableau Prep is a data preparation tool used to clean and organize data before analysis.

It's important because real-world data often has missing values, duplicates, or wrong formats, and Prep helps make it accurate and ready for visualization.

Steps:

1. **Remove Null Values:**
 - Select the field → Apply a filter → Exclude null entries.
2. **Remove Duplicate Records:**
 - Use the "Remove Duplicates" option or group data to keep unique rows.
3. **Change Incorrect Data Types:**
 - Click the field → Choose the correct data type (e.g., change String to Date or Number).

Q10. A retail manager wants answers to the following: Which category generates the highest profit? Which region performs poorly? How does sales change over time? a) Mention which charts you would use b) Identify one key insight that management could derive c) Explain how this helps in business decision-making

ANS- a) Charts to Use:

- Highest Profit by Category → **Bar Chart**
- Poorly Performing Region → **Map Chart**
- Sales Change Over Time → **Line Chart**

b) One Key Insight:

- Example: *Technology generates the highest profit, while the South region performs poorly.*
- The line chart may show that sales peak during certain months (seasonal trend).

c) Business Decision-Making:

- These insights help managers **focus on profitable categories**, improve weak regions, and plan inventory/marketing around sales trends.
- Result: Better resource allocation, targeted promotions, and improved overall performance.